

Refining indices of risk from salmonid hatcheries

Matthew R. Falcy*

U.S. Geological Survey, Idaho Cooperative Fish and Wildlife Research Unit, Department of Fish and Wildlife Sciences,
University of Idaho, Moscow, Idaho, USA

*Corresponding author: Matthew R. Falcy. Email: mfalcy@uidaho.edu.

ABSTRACT

Objectives: The risk to wild populations from domestication selection in hatcheries is commonly indexed with estimates of the proportion of hatchery-origin spawners (*pHOS*) on natural spawning grounds. Widely used calculations of *pHOS* are not commensurate with the genetic theory that motivates the use of *pHOS* as an index of risk from domestication selection.

Methods: I derive an alternative metric of *pHOS* that is commensurate with the spatiotemporal variability of natural- and hatchery-origin fish and the genetic theory used to index risk. I describe nuances that are associated with indices of hatchery risk, including how to calculate variances.

Results: The widely used calculation of *pHOS* will overestimate the risk when the densities of breeding natural- and hatchery-origin fish differ in space or time. The analytics provided here will improve the empirical estimation of an index of risk from hatchery-origin fish and the associated uncertainty.

Conclusions: Clarifying the calculations and interpretations of indices of risk to natural-origin fish can help managers balance multiple management goals associated with hatchery production.

KEYWORDS: domestication, hatchery, *pHOS*, risk, salmon

LAY SUMMARY

The widely used empirical metric of risk to natural salmonid populations from hatchery domestication selection, *pHOS*, is not calculated commensurately with theory. A calculation that is useful when there is assortative mating in space and/or time is provided.

INTRODUCTION

Fish hatcheries can provide many benefits to people and ecosystems, but there is also concern about their negative effects on the productivity of wild populations arising from domestication selection in hatcheries (Araki et al., 2009; Chilcote et al., 2011; McMillan et al., 2023). Estimating the historical and contemporary effects of domestication selection is difficult, so managers often rely on more easily measured metrics to index hatchery risk. The proportion of hatchery-origin spawners (*pHOS*) is an empirical metric that is used to index demographic and genetic risk to populations in the wild (Paquet et al., 2011). The metric *pHOS* is used to index genetic risk from domestication selection by indexing interbreeding between hatchery- and natural-origin fish. I contend that common calculations of *pHOS* do not properly capture the second index—the probability of interbreeding. I focus on the usefulness of *pHOS* as an index of genetic risk from domestication selection, but outbreeding depression and demographic risk is also considered.

Ford (2002) developed a quantitative trait genetic model of domestication selection (hereafter, “the Ford model”) and demonstrated its application to Pacific salmon *Oncorhynchus* spp. populations. The Ford model shows that the long-term equilibrium phenotype of individuals breeding in the wild depends, in part, on the proportions of hatchery- and natural-origin fish breeding in natural and captive environments (Figure 1A). A Gaussian function is used to convert equilibrium phenotype into equilibrium fitness. Fitness improves when *pHOS* declines and the proportion of natural-origin broodstock (*pNOB*) increases (Figure 1B). This result has led to widespread adoption of *pHOS* and *pNOB* as indices of risk from domestication selection over the past two decades. Indeed, these metrics are routinely reported in hatcheries’ genetic management plans of Pacific salmonids that are used to inform biological opinions under the Endangered Species Act.

It is difficult to measure *pHOS* empirically and more difficult still to do it in a way that is commensurate with the Ford model.

Received: November 9, 2024. Revised: September 24, 2025. Accepted: October 4, 2025

Published by Oxford University Press on behalf of American Fisheries Society 2026. This work is written by (a) US Government employee(s) and is in the public domain in the US.

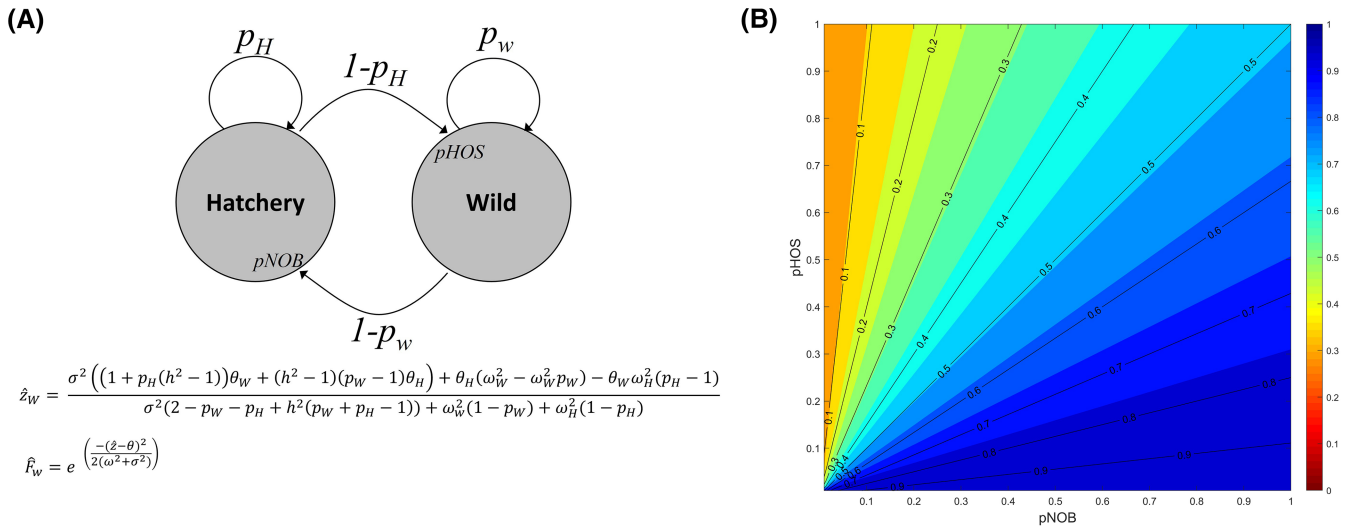


Figure 1. (A) The Ford (2002) model of domestication selection shows that the long-term equilibrium phenotypic value in the wild $\hat{z}_W$ depends on heritability (h^2), phenotypic variation among individuals (σ^2), and the proportions (p) of individuals moving between hatchery (H) and wild (W) environments. Ford (2002) uses a Gaussian function to convert equilibrium phenotypes into equilibrium fitness of animals breeding in the wild (F_W). The (B, colors) resulting equilibrium fitness of animals breeding in the wild improves with increasing proportion of natural-origin broodstock ($pNOB$) and decreasing proportion of hatchery-origin spawners ($pHOS$). The black isolines in panel B are $PNI = pNOB / (pNOB + pHOS)$, which can be seen to align with modeled fitness (colors) (PNI = proportionate natural influence). The parameter values that are used in panel B match the Ford (2002) scenario of “weak selection.”

Unlike $pNOB$, which is measured in the controlled environment of a hatchery and can thus be known with negligible error, measuring $pHOS$ usually involves intensive fieldwork. Let H and W represent the abundances of hatchery- and natural-origin fish, respectively, that spawn in the wild. Estimating H and W at the population scale leads to the widely used point estimator

$$pHOS = \frac{H}{H + W}. \quad (1)$$

I contend that this estimator is not commensurate with the Ford model when H and W do not exhibit panmixia. The Ford model presumes panmixia, but departures from panmixia can be expected to influence the magnitude of the genetic risk that is posed by hatchery-origin fish. For an extreme example, a single population that is composed of 1,000 H breeding near a hatchery and 1,000 W breeding farther upstream would have a $pHOS$ of 0.5. The effect of domestication selection would be very different from that of an alternative population with a $pHOS$ of 0.5 in which 1,000 H and 1,000 W freely interbreed with one another, as per the panmixia assumption in the Ford model.

Seemingly co-occurring mixtures of H and W Pacific salmonids often do not exhibit panmixia (Hoffnagle et al., 2008; Mackey et al., 2001). It is well known that the return timing of H fish can be manipulated to minimize breeding with W fish and provide additional fishing opportunities (Tillotson et al., 2019). Spatial segregation between H and W fish is also common. Indeed, areas around some hatcheries have been given special designations (“hatchery hot spot”) and unique $pHOS$ tolerances in planning documents (Oregon Department of Fish and Wildlife, 2014). Here, I present an alternative calculation of $pHOS$ that incorporates spatiotemporal clustering of H and W fish in a way that is commensurate with the Ford model. The

alternative calculation of $pHOS$ does not intend to measure all forms of risk from hatchery-origin fish. The demographic effects of displacement from high-quality spawning grounds and the density-dependent survival of juvenile progeny in estuaries contribute to the ecological risk that is posed by hatchery fish spawning in the wild (Kostow, 2009). For ecological risk, the traditional $pHOS$ metric (Equation 1) is useful, but variance estimators have not been developed until now.

METHODS AND RESULTS

I propose a population-level metric that is commensurate with the Ford model using W as weights in subunits of space and time. Assume that the raw data are unbiased counts of H and W spawners that are made over $i = 1, 2, 3, \dots$ I survey sites within a given population. Each survey site is resampled over $t = 1, 2, 3, \dots$ T occasions throughout the spawning season. At the scale of the entire population and spawning year, a metric that is commensurate with the Ford model is

$$pHOS_W = \sum_{i=1}^I \sum_{t=1}^T \left(\frac{H_{i,t}}{H_{i,t} + W_{i,t}} \right) \left(\frac{W_{i,t}}{\sum_i \sum_t W_{i,t}} \right), \quad (2)$$

where the subscript in $pHOS_W$ denotes weighting by abundances of natural-origin fish. The first parentheses contain the simple proportion of H fish for a sampling site and occasion. This fraction is multiplied by the proportion of W fish in the site and occasion relative to all W fish across all sampling locations and occasions within the population-year. The resulting product is then summed over all sampling locations and occasions. Equation 2 assumes that counts H and W are proportional to the relative abundances across sites and times.

Table 1. Hypothetical example of heterogeneity in abundances of hatchery- and natural-origin spawners across two sites within the same population; $pHOS_i$ = the proportion of hatchery fish within a site.

	Origin		$pHOS_i$
	Hatchery	Natural	
Site $i = 1$	100	500	0.167
Site $i = 2$	200	20	0.909
Sums	300	520	

A simple hypothetical example is provided in Table 1, where counts of hatchery- and natural-origin salmonids are made at just two sites within a population and there is no temporal component to the counts. Note that $pHOS_i$ (the proportion of hatchery fish within a site) is very different in the two sites. Also note that the two sites have very different numbers of natural-origin fish. A spatially naïve estimate is

$$pHOS = \frac{300}{300 + 520} = 0.366.$$

The recommended calculation is

$$pHOS_W = 0.167 \left(\frac{500}{520} \right) + 0.909 \left(\frac{20}{520} \right) = 0.195.$$

Note that in an extreme case with a single time step and two spatial units with complete segregation between H and W fish, $pHOS_W = 0$ because there is a 0 in the numerator of the second parentheses of Equation 2. This is reasonable because there is no genetic exchange in this scenario and thus no genetic risk. Additionally, if the proportion of hatchery-origin spawners is identical between the two spatial units, then $pHOS_W$ is equal to the simple and widely used metric that is spatially and temporally naïve:

$$pHOS = \frac{\sum_{i=1}^2 H_i}{\sum_{i=1}^2 H_i + W_i} \quad (3)$$

Note that the second parentheses in Equation 2 is a number (weight) in the interval $[0,1]$. This number is then multiplied into the widely used metric that is spatially and temporally naïve. Thus, $pHOS_W < pHOS$ whenever H and W fish do not exhibit panmixia because $pHOS \times W < pHOS$ when $W < 1$. The extent to which $pHOS_W < pHOS$ can be explored by drawing 1 million random uniform vectors of 10 probabilities that sum to 1 using the Dirichlet distribution and then using these vectors to assign 1,000 individuals to five spatial locations and two origin types (H and W ; Figure 2).

Differences in individual fitness may be explained, at least in part, by differences in fecundity that are associated with body size (Malick et al., 2023). In many sampling situations, the sex and age composition of spawners is known. It is possible to incorporate this information into an estimate of $pHOS_W$ (Hatchery Scientific Review Group, 2017, p. 49). Consider six age-classes, $a = 1, 2, 3, \dots, 6$. Let $F_{a,sex}$ be the fitness at age a for female or male fish.

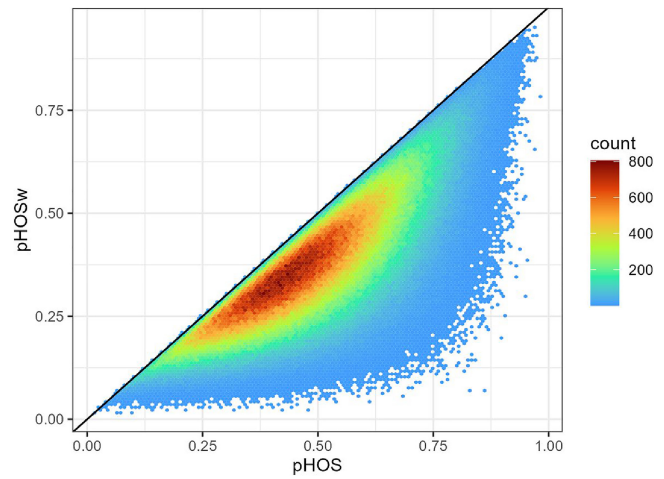


Figure 2. The metric $pHOS_W$ (the proportion of hatchery fish weighted by proportions of natural-origin fish) is less than the traditional metric proportion of hatchery-origin spawners ($pHOS$) for all 1 million random partitions of 1,000 individuals into five spatial groups and two origin types (hatchery and natural). The diagonal black line gives the 1:1 relationship. A Dirichlet distribution parameterized with a vector of 10 ones was used to randomize the partitioning of individuals into 10 groups.

$$pHOS_{i,t,female} = \frac{\sum_{a=1}^6 (H_{i,t,a,female} \cdot F_{a,female})}{\sum_{a=1}^6 (H_{i,t,a,female} \cdot F_{a,female} + W_{i,t,a,female} \cdot F_{a,female})}$$

$$pHOS_{i,t,male} = \frac{\sum_{a=1}^6 (H_{i,t,a,male} \cdot F_{a,male})}{\sum_{a=1}^6 (H_{i,t,a,male} \cdot F_{a,male} + W_{i,t,a,male} \cdot F_{a,male})} \quad (4)$$

$$pHOS_W = \sum_{i=1}^I \sum_{t=1}^T \left(\frac{pHOS_{i,t,female} + pHOS_{i,t,male}}{2} \right) \left(\frac{\sum_a (W_{i,t,a,female} \cdot F_{a,female} + W_{i,t,a,male} \cdot F_{a,male})}{\sum_i \sum_t \sum_a (W_{i,t,a,female} \cdot F_{a,female} + W_{i,t,a,male} \cdot F_{a,male})} \right)$$

Several things are noteworthy from here. First, $F_{a,sex}$ is unitless. The six values of F_a should just index the relative fitness at age. For example, in Table 2, fitness is highest for 5- and 6-year-old fish and declines with younger fish. Furthermore, fitness differences between male and female can be expressed as differences between $F_{a,female}$ and $F_{a,male}$. In Table 2, 2-year-old males have less fitness than 2-year-old females ($0.1 < 0.5$). However, as the fish age, the fitness differences between sexes disappear ($1 = 1$).

Variance of $pHOS_W$ and $pHOS$

The calculations for $pHOS_W$ given above explicitly acknowledge that proportions of hatchery-origin fish on the spawning grounds vary across space within a population and temporally within a year. Uncertainty in an estimate of $pHOS_W$ and $pHOS$ for a single population and year could inform risk-adverse management decisions and help guide the allocation of sampling effort across multiple monitoring objectives. Components of uncertainty in $pHOS_W$ and $pHOS$ are provided below, but details will depend on data collection protocols.

Table 2. Hypothetical example of sex- and age-specific differences in fitness (F).

Age a	$F_{a, \text{female}}$	$F_{a, \text{male}}$
1	0	0
2	0.5	0.1
3	0.7	0.5
4	0.8	0.8
5	1	1
6	1	1

Within site and occasion sampling variance

If all fish within a survey site and occasion are observed, then there is no uncertainty in the calculated proportion of hatchery-origin fish for that site and occasion. However, if a sample is taken from a larger statistical population of fish at a location and time, then the calculated proportion of hatchery fish is random. Let the count $H_{i,t}$ be given by the hypergeometric distribution. This is relevant to scenarios where sampling without replacement is performed. Let n = number of observed fish, N = total number of fish at the site on the occasion, and K = true number of hatchery fish at the site on the occasion. The variance of the observed proportion of hatchery-origin fish ($H_{i,t}/[H_{i,t} + W_{i,t}]$) due to random sampling is

$$\text{Var}(pHOS_{i,t}) = \frac{K(N-K)(N-n)}{nN^2(N-1)}. \quad (5)$$

This estimator is the variance of the hypergeometric random variable divided by n^2 . Dividing by n^2 follows from the scaling property of variances,

$$\begin{aligned} \text{Var}(aX) &= E[(aX)^2] - (E[aX])^2 = E[a^2X^2] - (aE[X])^2 \\ &= a^2E[X^2] - a^2(E[X])^2 \\ &= a^2(E[X^2] - (E[X])^2) = a^2\text{Var}(X), \end{aligned}$$

where X is the observed count, H , and a is $1/n$. The accuracy of this variance estimator can be checked with R code provided in [Appendix A](#). It may be reasonable to assume that all the fish that are present are detected ($n=N$), in which case $\text{Var}(pHOS_{i,t}) = 0$. Because K and N are never known empirically, professional judgement or more focused mark-recapture studies using Jolly-Seber models may be needed to estimate abundance and determine the proportion of available fish at a site and time that are observed by surveyors.

Among site and occasion sampling variance

If the goal is to estimate the variance associated with the mean of several $pHOS_{i,t}$, then the variances of the individual $pHOS_{i,t}$ ([Equation 5](#)) and their covariances must be considered. For nine individual $pHOS_{i,t}$ measures, the variance of their average is

$$\begin{aligned} \text{var}(\overline{pHOS}) &= \frac{1}{9^2} (\text{var}[pHOS_1] + \text{var}[pHOS_2] + \dots + \text{var}[pHOS_9] \\ &\quad + 2\text{cov}[pHOS_1, pHOS_2] + 2\text{cov}[pHOS_1, pHOS_3] \\ &\quad + \dots + 2\text{cov}[pHOS_8, pHOS_9]) \end{aligned} \quad (6)$$

This expression follows from the theory of moments of linear combinations of random variables, where the following has been proved:

$$\text{var}(Y) = \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i) + 2 \sum_{i < j} \sum a_i a_j \cdot \text{cov}(X_i X_j),$$

Note that [Equation 6](#) includes all the covariance terms for all the pairwise combinations of $pHOS_{i,t}$. With $M=9$ individual $pHOS_{i,t}$, there are $M!/(2! \times [M-2]!) = 36$ pairwise covariance terms. The values for $pHOS_{i,t}$ are likely to be highly correlated in space and time, so including the covariance is important. Estimating the spatial and temporal correlation among $pHOS_{i,t}$ could proceed with a variogram that is constructed from past data.

[Equation 6](#) is suitable for unweighted $pHOS$ ([Equation 1](#)). A small adjustment is needed to accommodate the weights used in $pHOS_w$ ([Equation 2](#)). A general formula for the variance of a weighted mean is

$$\text{Var}(\overline{pHOS}_w) = \sum_{i=1}^n \sum_{j=1}^n \Sigma_{ij} w_i w_j = w^T \Sigma w, \quad (7)$$

where Σ is the covariance matrix and w are the weights $\Sigma_{i=1}^n \Sigma_{j=1}^n \left[\frac{w_i w_j}{\Sigma_{i=1}^n \Sigma_{j=1}^n w_i w_j} \right]$ and T is the matrix transpose operator. This assumes that the weights are fixed. From here, there are several ways of arriving at a confidence interval. The simple Wald interval, $\theta \pm 1.96 \times \text{SE}$, is easy to compute but can return intervals that are negative when θ (i.e., $\overline{pHOS}_w$) is small. This is undesirable because $pHOS_w$ is a proportion that is bounded to the interval $[0,1]$. To avoid negative intervals, we can use the delta method on a logit scale,

$$\text{var} \left[\log \left(\frac{\theta}{1-\theta} \right) \right] \approx \frac{d}{d\theta} \left[\log \left(\frac{\theta}{1-\theta} \right) \right]^2 \text{var}(\theta) = \left[\frac{1}{\theta(1-\theta)} \right]^2 \text{var}(\theta), \quad (8)$$

and then back-transform using the inverse logit function. The R code to implement [Equations 2, 5, 6, 7, and 8](#) is in [Appendix B](#). The code generates 95% confidence intervals that include both within site/time variance ([Equation 5](#)) as well as the among site/time variance ([Equations 6–8](#)).

Statistical power

The variance estimator given in [Equation 6](#) does not address the effect of sample size on inferences about $pHOS_w$ and $pHOS$ at the population scale. As $pHOS_w$ and $pHOS$ are proportions that are bounded between 0 and 1, we may regard them as the result of a beta random variable. Maximum likelihood estimates of the parameters of the beta distribution can be made from the (correlated) observations of $pHOS_{i,t}$:

$$f(pHOS_{i,t} | a, b) = \frac{1}{B(a, b)} pHOS_{i,t}^{a-1} (1 - pHOS_{i,t})^{b-1},$$

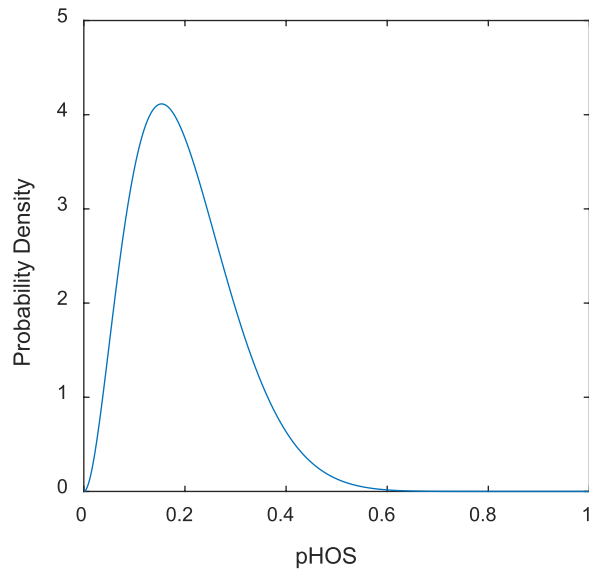


Figure 3. The beta distribution parameterized with mean = 0.2 and standard deviation = 0.1. The beta distribution can model variation among individual point estimates of $pHOS_{i,t}$ (the proportion of hatchery fish at site i and time t) within a breeding population; $pHOS$ = proportion of hatchery-origin spawners.

where B is the beta function, $B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$, and Γ is the gamma function, $\Gamma(X+1) = X! = \prod_{i=1}^X x$.

Assume that the true distribution of $pHOS_{i,t}$ within a population on a given run year has mean $\mu = 0.2$ and standard deviation $\sigma = 0.1$. Using method of moments, the associated beta distribution has parameters $a = 3$, and $b = 12$ because

$$a = \mu \left(\frac{\mu(1-\mu)}{\sigma^2} - 1 \right)$$

$$b = (1-\mu) \left(\frac{\mu(1-\mu)}{\sigma^2} - 1 \right).$$

In this case, where $a = 3$ and $b = 12$, the population-wide $pHOS_{i,t}$ generating function is given in Figure 3.

Suppose we have the following five random observations of $pHOS_{i,t}$ arising from the beta distribution given in Figure 3: 0.053, 0.255, 0.217, 0.241, 0.091. The maximum likelihood point estimates of a and b as well as the associated 95% confidence intervals are given in Table 3. Although the point estimates happen to be close to the truth ($a = 3$, $b = 12$), the confidence intervals are huge. Repeating this exercise with 50 observations yields much narrower confidence intervals (Table 3).

DISCUSSION

Simple calculations of $pHOS$ (Equation 1) that ignore assortative mating of hatchery- and natural-origin fish in space and time are not commensurate with the theory that is used to interpret $pHOS$ as a measure of risk from domestication selection. Hatchery- and natural-origin salmonids often exhibit fine-scale spatial and temporal heterogeneity in relative abundances while

Table 3. The effect of sample size on the accuracy of estimates of shape parameters (a and b) of the beta distribution. The confidence interval calculations are given in Appendix C; L95% = lower 95% confidence limit; U95% = upper 95% confidence limit.

Parameter	Sample size	Point estimate	L95%	U95%
$a = 3$	5	2.77	0.12	62.34
$b = 12$	5	13.5	0.23	790.68
$a = 3$	50	3.27	2.17	8.43
$b = 12$	50	12.93	4.94	19.84

they are breeding. Empirically capturing fine-scale heterogeneity is feasible with common field protocols. Weighting site- and time-level measures of $pHOS$ by the number of natural-origin fish yields a metric that is commensurate with models by Ford (2002) and the Hatchery Scientific Review Group (2004). The metric $pHOS_w$ (Equations 2 and 4) is a better index of risk from domestication selection because it is a more accurate proxy measure of gene flow into the natural-origin subpopulation than $pHOS$. Although fine spatial and temporal scales of enumeration are generally preferable, the usefulness of $pHOS_w$ will diminish if the scale of observation is so fine that fish are migrating rather than actively breeding. Further work is needed to develop recommendations around spatial and temporal scales that are too fine to confidently capture breeding distribution. Future recommendations could address (1) salmonids such as spring Chinook Salmon *Oncorhynchus tshawytscha* and summer steelhead *O. mykiss* that return to freshwater months prior to spawning, (2) surveys of live fish versus carcasses, and (3) itinerant polygamy (Bentzen et al., 2001). Most systems will have high uncertainty about the spawning disposition of individuals within a putative population. Additional simulations could explore the sensitivity of $pHOS_w$ to violations of the assumptions about the spatial and temporal units where spawning is presumed.

Although accounting for sex- and age-specific differences in fitness (Table 2) may be empirically difficult, it should be noted that ignoring potential differences among sexes and ages of fish will result in a computation of $pHOS_w$ that tacitly assumes homogeneity among all these fish. This means that failure to incorporate $F_{a,sex}$ is exactly equivalent to assuming $F_{a,sex} = 1$ for all ages and sexes. Future work could help create reporting standards for $F_{a,sex}$ when data are not available. Note that an attempt to incorporate fitness differences between hatchery- and natural-origin fish into $pHOS_w$ will “double-count” the effect because this is precisely what the Ford model addresses.

The methods that were developed here could be used to more thoroughly assess the effects of sample size on inference. For example, Equation 5 can be used to understand how uncertainty in $pHOS_{i,t}$ changes as surveyors’ ability to detect fish at a site and time changes. Additionally, a single realization from a beta distribution was used in Table 3, but repeated sampling could be used to characterize confidence interval coverage as a function of sample size. However, real $pHOS_{i,t}$ measures will likely be autocorrelated but the calculation of confidence intervals in Table 3 assumes that they are independent.

Not all genetic risk from hatcheries arises from domestication selection. A hatchery program that originated from out-of-basin broodstock with $pHOS_w = 0$ (because of breeding segregation in space or time) can still pose genetic risk.

Hatchery-origin parents (parental generation) that spawn in the wild will produce unmarked, natural-origin progeny (F_1 generation) that could subsequently breed with fish from the focal basin's natural-origin lineage. Their progeny (F_2 generation) could experience reduced fecundity due to outbreeding depression even though the F_1 parents are natural-origin fish. Although fitness loss may seem negligible because F_1 parents were able to complete a life cycle in the wild, the fitness consequences are not fully manifest until outbreeding depression manifests in the next generation. Thus, out-of-basin broodstock that interbreed with natural-origin spawners can reduce fitness in the wild due to phenotype/environment mismatch arising from influences by both the hatchery environment (domestication selection) and the exogenous basin (outbreeding depression). Future work could adapt the Ford model to accommodate out-of-basin effects by modeling two traits: one trait under domestication selection with two-way gene flow and another trait that evolves from the optimum in the exogenous basin toward the focal basin's optimum without back-migration to the exogenous basin, and with or without continuous immigration to the focal basin from the exogenous basin.

There is often confusion about transgenerational effects from hatchery domestication selection. Ford (2002) showed a model of fitness that can be iterated generationally, but the Ford model underlying management recommendations for $pHOS$ and $pNOB$ are based on Ford's (2002) functions for the long-term equilibrium mean fitness resulting from continued interbreeding between natural- and hatchery-origin subpopulations (Figure 1). Because recommendations concerning $pHOS$ and $pNOB$ (Figure 1B) are based on long-term equilibrium mean fitness for fixed $pHOS$ and $pNOB$, there is usually no need for additional attempts to account for multigenerational effects outside of the Ford model. However, rates of interbreeding are not constant through time. $pHOS_w$ measures current-year interbreeding at the places and times fish actually meet. An assessment of second-generation effects wherein natural-origin offspring of hatchery parents (F_1) later breed with wild fish under a different level of $pHOS_w$ could be assessed by iterating Ford's equations generationally. This leads to the possibility for dynamic management. In years when natural-origin spawner abundance is critically low, managers may elect not to take fish into the hatchery to avoid "mining" the last few natural-origin fish. This would reduce $pNOB$ for 1 year, setting up a "genetic debt" that could be repaid in the future by, for example, reducing $pHOS$ for one or more years. Future work could develop models to dynamically manage $pHOS_w$ and $pNOB$ to achieve long-term goals while accounting for transgenerational effects.

Monitoring $pHOS_w$ is a simple way of indexing risk from domestication selection. As demonstrated in Figure 1B, the metric proportionate natural influence (PNI) maps into fitness isoclines extremely well. However, when the strength of selection is greater, the relationship between PNI and fitness worsens (not shown). Future work that informs estimates of the strength of selection would help to quantify risk from domestication selection.

The relative importance of genetic risk and ecological risk may shift according to phases of recovery (Hatchery Scientific Review Group, 2020), requiring careful consideration of both $pHOS_w$ and $pHOS$. Decision support tools that incorporate

uncertainty in $pHOS_w$ and $pHOS$ could help managers balance the risk and rewards from hatcheries more explicitly. The work presented here is a step toward refining calculations of a metric of risk from hatchery-origin fish that has been used under the Endangered Species Act since broadscale hatchery reform efforts began approximately two decades ago.

DATA AVAILABILITY

No data were used in this article.

ETHICS STATEMENT

This article meets the ethical guidelines and legal requirements set forth by the U.S. Geological Survey.

FUNDING

None declared.

CONFLICTS OF INTERESTS

None declared.

ACKNOWLEDGMENTS

Three anonymous reviewers provided commentary that significantly improved this manuscript. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

REFERENCES

- Araki, H., Cooper, B., & Blouin, M. S. (2009). Carry-over effect of captive breeding reduces reproductive fitness of wild-born descendants in the wild. *Biology Letters*, 5, 621–624. <https://doi.org/10.1098/rsbl.2009.0315>
- Bentzen, P., Olsen, J. B., McLean, J. E., Seamons, T. R., & Quinn, T. P. (2001). Kinship analysis of Pacific salmon: Insights into mating, homing, and timing of reproduction. *The Journal of Heredity*, 92, 127–136. <https://doi.org/10.1093/jhered/92.2.127>
- Chilcote, M. W., Goodson, K. W., & Falcy, M. R. (2011). Reduced recruitment performance in natural populations of anadromous salmonids associated with hatchery-reared fish. *Canadian Journal of Fisheries and Aquatic Sciences: Journal Canadien Des Sciences Halieutiques Et Aquatiques*, 68, 511–522. <https://doi.org/10.1139/F10-168>
- Ford, M. J. (2002). Selection in captivity during supportive breeding may reduce fitness in the wild. *Conservation Biology: The Journal of the Society for Conservation Biology*, 16, 815–825. <https://doi.org/10.1046/j.1523-1739.2002.00257.x>
- HatcheryScientific Review Group. (2004). White paper no. 1: Predicted fitness effects of interbreeding between hatchery and natural populations of Pacific salmon and steelhead. In *Columbia River Hatchery Reform Project final systemwide report* (Appendix A). https://www.streamnet.org/app/hsrc/docs/CRB_4_appendix_a_white_papers.pdf
- Hatchery Scientific Review Group. (2017). *Implementation of hatchery reform in the context of recovery planning using the AHA/ISIT tool*. <https://www.streamnet.org/home/data-maps/hatchery-reform/hsrc-reports/>
- Hatchery Scientific Review Group. (2020). *Developing recovery objectives and phase triggers for salmonid populations*. <https://www.streamnet.org/home/data-maps/hatchery-reform/hsrc-reports/>

- Hoffnagle, T. L., Carmichael, R. W., Frenyea, K. A., & Keniry, P. J. (2008). Run timing, spawn timing, and spawning distribution of hatchery- and natural-origin spring Chinook Salmon in the Imnaha River, Oregon. *North American Journal of Fisheries Management*, 28, 148–164. <https://doi.org/10.1577/M05-145.1>
- Kostow, K. (2009). Factors that contribute to the ecological risks of salmon and steelhead hatchery programs and some mitigating strategies. *Reviews in Fish Biology and Fisheries*, 19, 9–31. <https://doi.org/10.1007/s11160-008-9087-9>
- Mackey, G., McLean, J. E., & Quinn, T. P. (2001). Comparisons of run timing, spatial distribution, and length of wild and newly established hatchery populations of steelhead in Forks Creek, Washington. *North American Journal of Fisheries Management*, 21, 717–724. [https://doi.org/10.1577/1548-8675\(2001\)021<0717:CORTSD>2.0.CO;2](https://doi.org/10.1577/1548-8675(2001)021<0717:CORTSD>2.0.CO;2)
- Malick, M. J., Losee, J. P., Marston, G., Agha, M., Berejikian, B. A., Beckman, B. R., & Cooper, M. (2023). Fecundity trends of Chinook Salmon in the Pacific Northwest. *Fish and Fisheries*, 24, 454–465. <https://doi.org/10.1111/faf.12738>
- McMillan, J. R., Morrison, B., Chambers, N., Ruggerone, G., Bernatchez, L., Stanford, J., & Neville, H. (2023). A global synthesis of peer-reviewed research on the effects of hatchery salmonids on wild salmonids. *Fisheries Management and Ecology*, 30, 446–463. <https://doi.org/10.1111/fme.12643>
- Oregon Department of Fish and Wildlife. (2014). *Coastal multi-species conservation and management plan*. https://www.dfw.state.or.us/fish/crp/docs/coastal_multispecies/CMP_main_final.pdf
- Paquet, P. J., Flagg, T., Appleby, A., Barr, J., Blankenship, L., Campton, D., Delarm, M., Evelyn, T., Fast, D., Gislason, J., Kline, P., Maynard, D., Mobrand, L., Nandor, G., Seidel, P., & Smith, S. (2011). Hatcheries, conservation, and sustainable fisheries-Achieving multiple goals: Results of the Hatchery Scientific Review Group's Columbia River basin review. *Fisheries*, 36, 547–561. <https://doi.org/10.1080/03632415.2011.626661>
- Tillotson, M. D., Barnett, H. K., Bhuthimethee, M., Koehler, M. E., & Quinn, T. P. (2019). Artificial selection on reproductive timing in hatchery salmon drives a phenological shift and potential maladaptation to climate change. *Evolutionary Applications*, 12, 1344–1359. <https://doi.org/10.1111/eva.12730>

APPENDIX A: THIS SIMULATION IN R DEMONSTRATES THE ACCURACY OF EQUATION 5 IN THE MAIN TEXT

```
N<-70 #True number of fish (H+W)
K<-40 #True number of H fish
n<-25 #Number of fish observed
(K*(N - K)*(N - n))/(n*N^2*(N - 1)) #Theoretical variance of H/(H+W)

x<-rhyper(nn=100000,m=K, n=N-K, k=n)
var(x/n) #Empirical variance of H/(H+W)
```

APPENDIX B: R CODE TO COMBINE UNCERTAINTY WITHIN AND AMONG SITES/TIMES AND GENERATE A 95% CONFIDENCE INTERVAL

```
#pHOSw
Eq.2<-function(H,W){
  sum((H/(H+W))*(W/sum(W)))
}

#Variance within
Eq.5<-function(K,N,n){
  (K*(N-K)*(N-n))/(n*N^2*(N-1))
}

#Variance among
Eq.6<-function(Sigma){
  1/length(diag(Sigma))^2*(sum(diag(Sigma))+sum(2*Sigma[upper.tri(Sigma)]))
}

#Generate hypothetical data
obs<-10 #number of sites/locations sampled
set.seed(1)
N.i<-rpois(n=obs, lambda=80)#Total number of fish at 10 site/locations
p<-rbeta(n=obs,10,10)#proportion hatchery
K.i<-N.i*p#number hatchery
n.i<-c(rep(40,ceiling(obs/2)),rep(30,floor(obs/2)))#number of observed fish
H.i<-rhyper(nn=obs,m=K.i ,n=N.i-K.i ,k=n.i)#'white balls' in R are Hatchery fish here.
W.i<-n.i-H.i

#Build a hypothetical covariance matrix.
#This doesn't create covariances. It just puts 0.01 or 0.02 where empirical numbers belong.
#Must ensure covariance matrix is positive semidefinite.
Sigma.i <- matrix(data=0.01, nrow=obs, ncol=obs)
Sigma.i[1:ceiling(obs/2), 1:floor(obs/2)] <- 0.02
```

```

diag(Sigma.i) <- Eq.5(K=K.i,N=N.i,n=n.i)#Diagonal get variance within
pHOS<-Eq.2(H=H.i,W=W.i)
Var_pHOS<-Eq.6(Sigma.i)#This variance not advised for pHOSw, but fine for pHOS.
SE<-sqrt(Eq.6(Sigma.i))
lower95<-pHOS-1.96*SE
upper95<-pHOS+1.96*SE
c("pHOS"=pHOS,"Var"=Eq.6(Sigma.i), "SE"=SE,"Lower95"=lower95,"upper95"=upper95)
#Using weights, pHOSw
pHOSw<-Eq.2(H=H.i,W=W.i)
w <- W.i / sum(W.i)
Var_pHOSw <- as.numeric(t(w) %*% Sigma.i %*% w) #Equation 7
SE <- sqrt(Var_pHOSw)
lower95 <- pHOSw - 1.96 * SE
upper95 <- pHOSw + 1.96 * SE
c("pHOSw" = pHOSw, "Var" = Var_pHOSw, "SE" = SE,"Lower95" = lower95, "Upper95" = upper95)
#Now use delta method
#Prevent numerical problems with 0 or 1
eps <- 1e-6
pHOSw_adj <- pmin(pmax(pHOSw, eps), 1-eps)
# Equation 8.
logit_pHOSw <- log(pHOSw_adj/(1-pHOSw_adj))
Var_logit <- (1/(pHOSw_adj*(1-pHOSw_adj)))^2 * Var_pHOSw
SE_logit <- sqrt(Var_logit)
# 95% CI on logit scale
logit_lower <- logit_pHOSw - 1.96 * SE_logit
logit_upper <- logit_pHOSw + 1.96 * SE_logit
#Back transform with inverse logit
inv_logit <- function(x) exp(x)/(1+exp(x))
ci_lower_logit <- inv_logit(logit_lower)
ci_upper_logit <- inv_logit(logit_upper)
c("p_hat" = pHOSw, "Var" = Var_pHOSw, "SE" = sqrt(Var_pHOSw), "Lower95" = ci_lower_logit,
"Upper95" = ci_upper_logit)

```

APPENDIX C: R CODE TO CALCULATE CONFIDENCE INTERVALS IN [TABLE 3](#)

```

x<-rbeta(50,shape1=3,shape2=12)
fn<-function(x,params){
  NLL<- -sum(dbeta(x,params[1],params[2],log=T))
}

opt<-optim(par=c(1,1),fn=fn,x=x,hessian=T)
fisher.info<-solve(opt$hessian)
sig<-sqrt(diag(fisher.info))
upper<-opt$par+1.96*sig
lower<-opt$par-1.96*sig
data.frame(Point.Est=opt$par,Lower95=lower,Upper95=upper)

```